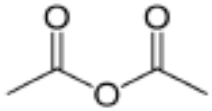
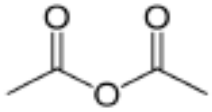
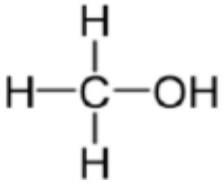
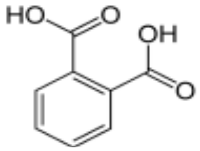
Compounds for derivatization formula, CAS #, purity, amount, type of packaging, price in US $	**Structure**	δ^{2}H (or δD) (mean value in ‰ vs. VSMOW, ± 1σ) (range) (# of measurements)	δ^{13}C (mean value in ‰ vs. VPDB-LSVEC, ± 1σ) (range) (# of measurements)
Acetic anhydride #1, $C_4H_6O_3$, CAS # 108-24-7, 99.5 %, ca. 1 mL sealed under argon in glass ampoule, US $250.		**-133.2** ± 2.1 ‰ from -131.5 to -136.0 ‰ n = 4	**-20.98** ± 0.03 ‰ from -20.94 to -21.01 ‰ n = 4
Acetic anhydride #2, $C_4H_6O_3$, CAS # 108-24-7, 99.5 %, ca. 1 mL sealed under argon in glass ampoule, US $250.		**-200.5** ± 1.5 ‰ from -198.5 to -202.5 ‰ n = 10	**-38.65** ± 0.01 ‰ from -38.64 to -38.65 ‰ n = 5
Methanol, CH_3OH, 99.8 %, anhydrous, CAS # 67-56-1, the δ^{2}H values characterize: (1) bulk hydrogen; (2) methyl hydrogen (calculated after subtracting the OH-hydrogen that was liberated in reactions between MeOH and Na metal). δ^{13}C was determined in bulk methanol. 5 mL sealed in glass ampoule, US $250.		**bulk methanol:** **-112.6** ± 0.8 ‰ from -111.8 to -113.5 ‰ n = 3 **methyl hydrogen:** **-141** ± 3 ‰ from -138 to -143 ‰ n = 3	**-46.77** ± 0.04 ‰ from -46.74 to -46.82 ‰ n = 3
Phthalic acid #2, $C_8H_6O_4$, CAS # 88-99-3, δ^{2}H measured in Na-phthalate to exclude carboxyl hydrogen. δ^{13}C measured in free acid. 3 g in glass vial, US $250		**-81.9** ± 1.2 ‰ from -81.8 to -83.0 ‰ n = 4	**-29.98** ± 0.01 ‰ from -29.96 to -29.99 ‰ n = 3